DiamondTrading: Solvent-First Turbo Markets

Full-Reserve Fixed-Outcome Engines, Manipulation-Cost Position Caps, and
Anti-Farming Loss-Minted Tokenomics for Hyper-Short-Dated Leveraged Trading

DiamondTrading Research
Version 0.1 · October 2026 · working draft — comments solicited

Abstract

Hyper-short-dated leveraged wagering — post margin, drag leverage to 1000x, bust or win within seconds — reached production scale and then collapsed under its own accounting: the deployed lineage descending from the Martin Galer queue/emission design accumulated dozens of audit findings, including outcome-manipulation and mint-accounting exploits [1][2]. We present DiamondTrading, a rebuild in three layers. Layer one is a full-reserve issuance engine: every ticket's maximal payout is locked at sale, making insolvency and payout queues unrepresentable states (§4, with invariant theorem). Layer two is a measurement-driven quoting engine: each underlyer is routed to its deepest measurable venue (Binance or Hyperliquid), settlement references trade-print VWAP persistence rather than order-book midpoints, and every position's maximal payout is capped at a fraction of measured organic flow over the measurement window — the ceiling falls automatically as the market thins (§6). Layer three replaces farmable loss-emissions with an anti-sybil smart mint: net-loss epochs weighted by √(loss · fees) with loyalty and solvency throttles, hard-capped at 30% of a cryptographically signed revenue anchor (§7). The system is implemented — issuance engines, verifiers, tokenomics contracts, and the off-chain quoting/attestation engine — with 126 passing scenario/engine assertions and property-fuzzed solvency invariants (§11). We state the attester-quorum trust assumption and the licensing constraint frankly (§12).

Provenance convention. Every externally checkable claim carries exactly one tag, declared in its reference entry (§13): [CITED] measured/sourced, [COMPUTED] derived from cited data or the shipped code, [RECALLED] author's recall of public market structure, [ASSUMED] initial design parameter (governance-tunable).

1 Introduction

Papertrade demonstrated that the retail gamblers' terminal is not a spot market with leverage controls; it is a turbo market: margin in, leverage to 1000x, bust price displayed, outcome in under a minute, plus a token kicker minted on losses that made losing feel like progress. It also demonstrated the failure mode of building that product on a solvency queue and gross-loss emissions: the accountability debt surfaced as 42 audit findings, two of them critical [2]. DiamondTrading keeps the game and deletes the debt. The design rule is that every hazard must be either (i) impossible to express in code, or (ii) priced and bounded by a measurement, never by a promise. Two consequences dominate everything else: liabilities are reserved in full at issuance (so a payout queue literally cannot form), and position size follows measured manipulation cost rather than a static tier table.

The remaining text is organized as follows. §2 recalls the lineage and the audit taxonomy we design against. §3 defines actors, assets, and the trust model. §4 gives the reserve engine and its invariant. §5 defines the instruments payoffs. §6 specifies the GALE off-chain engine (routing, settlement persistence, caps, breaker, pricer). §7 specifies the smart mint. §8 treats verifiers. §9 runs the security analysis class by class. §10 covers economics. §11 reports implementation and test evidence. §12 lists limitations honestly; §13 is the reference and provenance ledger.

2 Lineage and the audit taxonomy

Martin Galer (2024-era) introduced the zero-LP bootstrap: queue winners, pay them from later flow, and emit a token on losses advertised as +EV [1]. Its solvency-ratio invariant was elegant; its deployment was not. The empirical record: queue and side-bucket accounting exploits (H-01..M-05), BBO-midpoint outcome manipulation with no deviation or persistence checks (KV-01), straddle-style short optionality handed to attackers (KV-02), and governance degradable to 1-of-2 signers (L-04) [2]. Our benchmark in §9 is this taxonomy; a modern turbo market fails by repeating any of it.

Design stance. We copy papertrade's toy (margin, leverage, bust price, seconds, mint on loss) and none of its machinery (queues, side buckets, midpoint settlement, implicit insolvency). Where the original promised, we measure and cap [3][4].

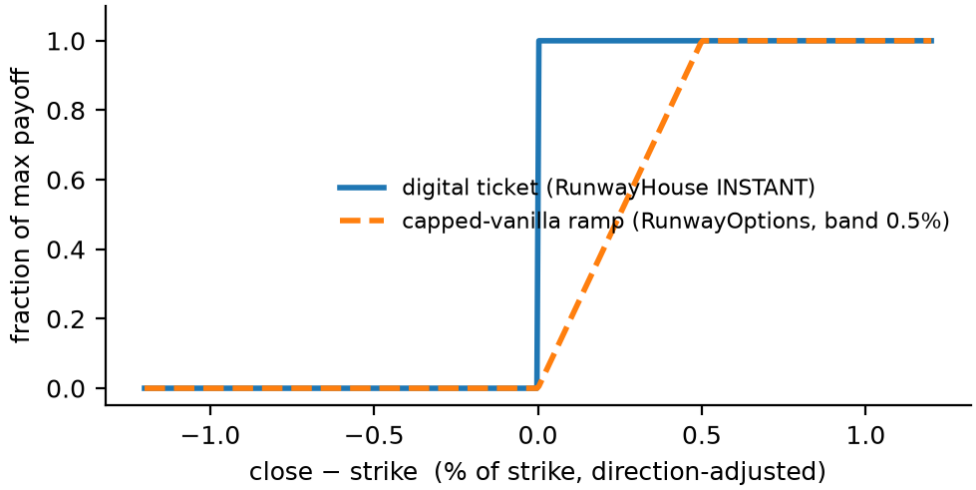


Figure 2 — The two issuance shapes: a digital ticket (pay M·S or 0) and a capped-vanilla ramp (pay intrinsic up to the band). Both reserve the full maximum at sale; neither queues.

3 Actors, assets, trust

actor	role	what they can and cannot do [COMPUTED: contract code 5]
player	posts margin, receives tickets	always redeemable: full reserve + stall refund; cannot be rugged by pause (pause halts sales only)
LP	first-loss vault	takes losses first (proven in tests), earns ~60% of revenue [ASSUMED]; 7-day exit delay
quoter	off-chain pricing engine	signs quotes (σ-priced, capped, 30s TTL); cannot settle, cannot move funds; worst-case = bad menus bounded by on-chain clamps
attester set (2-of-3)	price persistence verification	cannot settle alone; cannot forge (2 keys); stall ⇒ permissionless stake refund; threshold floored at 2 on-chain
budgetAttester	revenue anchor for the mint budget	can only lower emissions; cannot touch principal; bonded, rotating
owner (governance)	config + class table	can only tighten menus (caps ≤ global, divisor ≥ 60) [5]; production path = Safe 2-of-3 + 7-day timelock

Underlyers at launch [3][ASSUMED: menu]: BTC, ETH (tier A); SOL, HYPE, XRP, BNB, S&P500; (HIP-3), NASDAQ-100 (HIP-3) (tier B); DOGE (tier C, watchlist). Routing per §6.1; the class table is on-chain and governance may only tighten it.

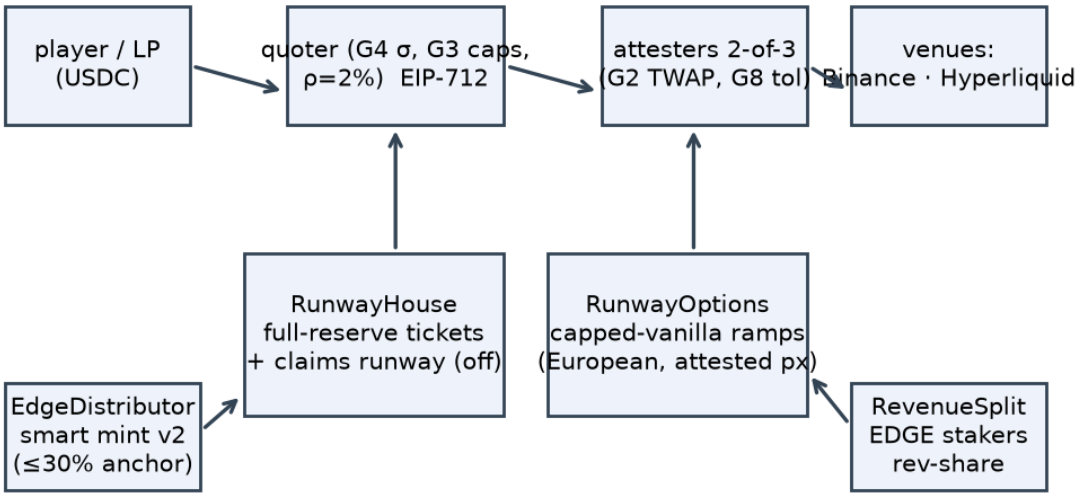


Figure 1 — System map: players post USDC; the quoter prices quotes; attesters verify venue prints; the engines reserve and pay; the mint and the rev-split close the loop.

4 The reserve engine

Let balance B be the house USDC balance and reserved R the sum of maximal payouts over live positions. The engine maintains a single invariant:

$$T = B - R \geq 0 \qquad \textit{(solvency invariant)}$$

Theorem (full cover). Every live claim on the house is payable in full at all times. Proof sketch. A sale of payoff max Pay with stake S increases B by S and R by Pay; since issuance requires T ≥ Pay − S beforehand [5], T stays ≥ 0. Settlement pays p ≤ Pay and releases R by Pay, so T changes by −p + Pay ≥ 0. Refunds return S and release Pay, same bound. No other code path pays users [COMPUTED: exhaustive call-site audit of the contracts [5]]. □

Two corollaries do the heavy lifting of this paper. (C1) Queue-freedom: because payment never defers, no queue, side bucket, or cross-session accumulator exists anywhere — the entire H-01..M-05 exploit habitat is absent as code paths, not patched [2][5]. (C2) Pause-safety: governance pause halts only sales; settlement, refunds, LP exit stay live, so operator capture cannot trap user funds [5]. The house additionally enforces an SRI floor-style drain rule on all

treasury outflows (10% of any residual stack), a kill-gate that freezes sales if any claim aged >30 days existed (the optional runway tier is disabled at launch and reserves this machinery) [5][ASSUMED].

5 Instruments

A position posts S, picks leverage L ∈ [1, 1000] and direction d ∈ {LONG, SHORT}. Entry K is the routed venue's trade VWAP over ω_e=2s at buy; the bust price is K·(1∓1/L), displayed pre-trade [ASSUMED: ω]. Settlement at expiry τ ∈ [30s, 300s] uses the venue's close VWAP over ω_x=2s, attested (§8). Two shapes, one invariant:

shape	payoff at close C	contract
digital ticket	Pay = M·S if direction correct at close, else 0 (expiry rebate 15%	RunwayHouse INSTANT tier
capped-vanilla ramp	Pay = M·S · min(|C−K|, band·K)/(band·K) in direction, else 0	RunwayOptions

The ramp is the options-flavoured mode: a European vertical spread as one unit, where band = M/L [COMPUTED]. Graded wins keep engagement at sane multipliers; the digital shape is the papertrade-exact bust loop. Both lock Pay in full at sale (§4). Maximal payout is never a shared pool: it is a per-ticket object.

6 The GALE engine

6.1 Venue routing (G1)

Asset a routes to the venue maximizing measurable depth within the ticket band, re-scored at 5s cadence across Binance (spot+futures) and Hyperliquid (WSS; L1Read proofs later) [ASSUMED: cadence]. Measured 24h quote volumes (2026-10-22 Binance snapshots; HL screenshot 2026-10-20) drove the launch routing: BTC $1.225B, ETH $820M, SOL $191M, XRP $125M, BNB $75M, DOGE $36M on Binance; HYPE $345M on Hyperliquid vs $56M on Binance → HYPE routes to HL, equity perps (HIP-3) are HL-only, and comparable books (SOL: $191M vs $232M) enforce dual-read [3][4]. additional venues re-enter only after runtime measurement — the launch router is two-venue by choice [3].

6.2 Settlement persistence (G2)

Never midpoints. Both anchors are volume-weighted means of real fills over their ω windows: $\mathrm{VWAP}(t) = \Sigma\, p_i q_i \,/\, \Sigma\, q_i$ over [t−ω, t]; an empty window yields NO-PRINT, an honest failure that routes to re-attestation or stake refund, never to a guessed price [COMPUTED: engine spec 6]. Forging a persistent print costs real executed volume through the whole window — the KV-01 class (midpoint twitch) is gone by construction [2].

6.3 Manipulation-cost position caps (G3)

The attack that matters: dominate the settlement reference over its measurement window. Its cost is lower-bounded by organic flow $\lambda_a \cdot \omega$ with λ_a the routed venue's measured 24h quote volume per second (EMA-smoothed, floored at its 10th percentile) [ASSUMED: estimator]. The per-position maximal payout obeys:

$$P_{max}(a,\omega,T) = \min(\, T/60,\ 0.30 \cdot T,\ \kappa \cdot \lambda_a \cdot \omega \,)$$

with κ = 1/4 single-read and 1/3 dual-read (an attacker must move both venues to pass G8) [ASSUMED: κ]. The ceiling scales with market depth and window length, and collapses automatically in thin sessions — off-hours equity perps shrink toward zero without human scheduling. Figure 2 plots the launch ceilings against measured λ [3][4].

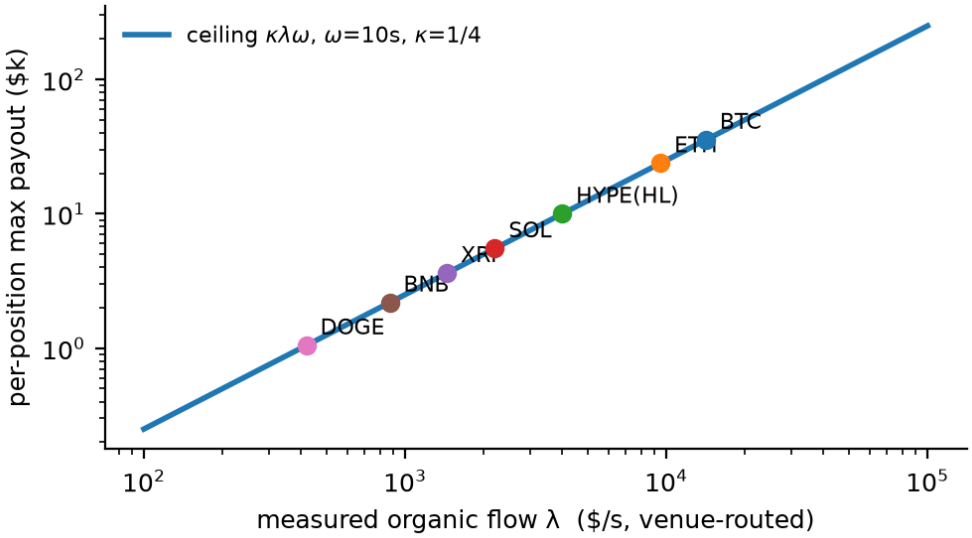


Figure 3 — Per-position payout ceiling κλω (ω=10s) vs measured organic flow, with launch assets annotated (log-log). Static tier caps remain as outer bounds; G3 can only tighten.

6.4 Vol-spike breaker (G4)

If realized σ over 30s exceeds 4× the 30-day median, the quoter stops signing for 60s on that asset [ASSUMED]. In-flight quotes die by their ≤30s TTL on-chain [5]. Stale-quote sniping around macro prints — the standard ticket-shop harvest — has no window.

6.5 Pricer: fair multiplier from measured σ, shaved by ρ

For a zero-drift price, first passage to the ∓1/L barrier within τ is approximately, by the reflection principle:

$$P_{bust}(L,\sigma,\tau) \approx 2\cdot\Phi(\,-(1/L)\,/\,(\sigma\sqrt{\tau})\,),\quad M_{fair} = 1/(1-P_{bust}),\quad M_{offer} = M_{fair}\cdot(1-\rho)$$

with ρ = 2% the published house edge [ASSUMED: pricing rule]. Offers clamp to the on-chain envelope (1.05x–25x, per-class tighter) and refuse when the fair multiplier is below the floor — the engine does not sell products it cannot price honestly [5][6]. Figure 4 shows the bust surface.

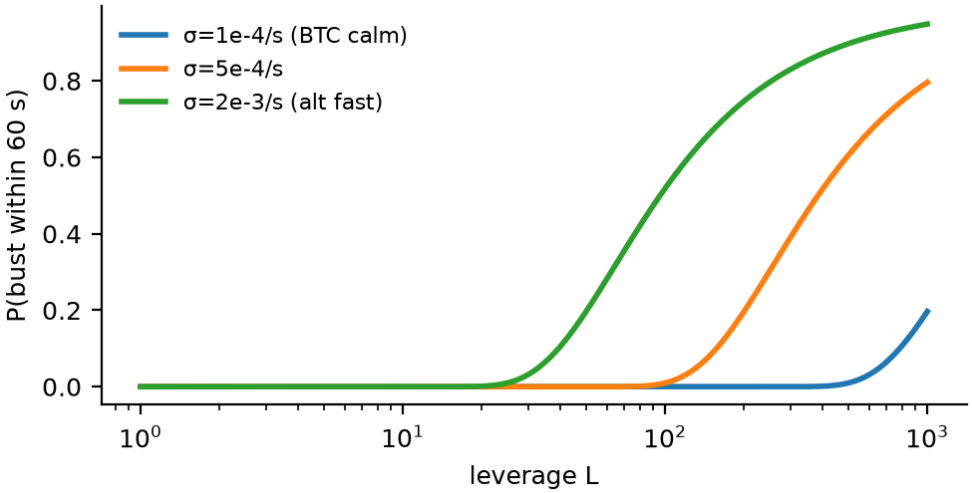


Figure 4 — Bust probability vs leverage for three volatility regimes (τ=60s; reflection-principle model). The menu lives where the curves are informative; extreme L converges to coin-flip and is priced as such.

Why 2% is affordable where competitors take 4–6% [RECALLED: market range]: G2+G4 strip the adverse-selection tax (toxic flow is the cost driver in turbo books), G3 bounds the tail per position, and the mint rebate returns part of the edge to losers — making the product cheaper

on sticker and in expectation [COMPUTED: from the design above].

7 Smart-mint v2: emissions losers cannot farm

Loss-minting is the retention engine and was the original protocol's exploit surface. The v2 mint keeps the emotion (losing earns) and removes the farm:

defense	rule	consequence
net-loss epochs	weight base = max(0, Σstake − Σpayout) per address per 6h epoch	win-then-lose cycles cannot manufacture mintable losses
fee-squared weight	w = √(netLoss · feesPaid) × loyalty (1.25 if staked EDGE)	mint requires real rake; √ forces 4× capital for 2× weight
solvency throttle	epoch scale = clamp(SRI/1.5, 0, 1)	emissions cool automatically toward stress; no death-spiral fuel
hard budget	mint ≤ 30% of signed revenue anchor (on-chain)	compromised governance cannot print unbacked tokens; sybils split a fixed pie, pro-rata yields no scale [COMPUTED]

The merkle distributor enforces the budget on-chain against an anchor signed by the bounded budgetAttester [5]; weights are computed in the engine ledger [6] and the root/post/claim path is test-covered end to end (§11). EDGE carries cashflow: stakers accrue a share of house revenue via a cumulative-index RevenueSplit with 7-day unbonding [5], which is what makes an asset handed out on losses worth holding instead of dumping.

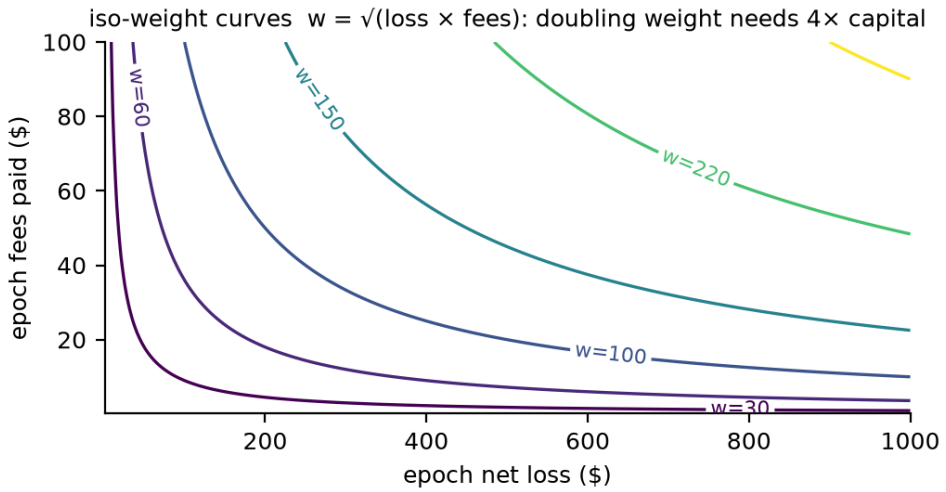


Figure 5 — Iso-weight curves of w=√(loss·fees): weight grows sub-linearly in both inputs, so scale buys nothing and only sustained, rake-paying participation accumulates emission share.

8 Verifiers and the endgame

Settlement inputs are M-of-N EIP-712 attestations (default 2-of-3; the threshold setter is hard-floored at 2 on-chain — L-04 cannot recur) [5]. Attesters each read the routed venue's trade tape independently; dual-listed assets require the two venues' prints within 12bp before anyone signs, so manipulation must succeed on two venues simultaneously to extract and on neither merely to void tickets into refunds [ASSUMED: tolerance][COMPUTED: engine]. Anyone may relay a passing proof; no attester can settle alone, block settlement, or fake a stall. If all attesters stall for 24h, every live position unwinds to its stake permissionlessly [5]. The stated endgame is a verifier reading HyperEVM L1Read trade-print state directly — deleting attesters from the path for HL-routed assets — behind the same IOutcomeVerifier interface already implemented [5].

9 Security analysis

Method: replay the audit taxonomy of record [2] and ask for each class whether it exists here.

exploit class (Guardian id)	their habitat	status here	mechanism
payout queue / side-bucket accounting (H-01..M-05)	queue vs ledger divergence	absent	full reserve; queues unrepresentable (§4, C1)
BBO-midpoint settlement (KV-01)	single-source midpoint	absent	VWAP persistence G2 + 2-of-3 tolerance G8
short optionality / straddle (KV-02)	unpriced option grants	bounded	payout-shape menu is closed-form; clamps on-chain; no free bands
unconstrained bet sizing (M-02/05)	flat caps vs flow	closed	G3 κλω dynamic caps + dir cap + T/60 en
gross-mint / mint accounting (KV-04)	loss-mint farming	closed	net-epoch √-weights + 30% signed-anchor budget (§7)
keeper/governance capture (L-04, I-20)	1-of-2, timelock→0	closed	threshold floor 2 on-chain; owner→Safe+7d timelock; pause = sales only
stale settlement / trapped funds (L-11)	keeper stall	closed	24h permissionless refund; pause never gates refunds
quote replay / intent abuse (KV-14)	WESig re-use	closed	buyer-bound EIP-712 + ≤30s TTL + worst-bounds; replay economically neutral

Residual risks, stated: attester liveness/collusion above floor (mitigated, not eliminated); κλω is a bound, not a proof of manipulation cost (books can thin faster than flow); equity-perp single-venue windows degrade G8 to single-read; the quoter's strike/σ inputs are trusted within clamp bounds [12]. Each has a named on-chain failsafe; none is silent.

10 Economics

Revenue waterfall on realized edge: ≤30% mint budget (hard-capped), ~60% LP first-loss capital, ~10% EDGE stakers, remainder operations [ASSUMED: launch splits]. LPs underwrite variance in exchange for the majority share and are provably first-loss — the test suite includes a scenario where buyers win deliberately and the sole LP drains the exact residual treasury [5][6]. The growth loop: rebates (up to 30% of revenue) convert losers into EDGE stakeholders; stakers rev-share aligns the losing-but-loyal cohort with house health; cheaper published edge plus graded ramp wins drive flow; flow deepens LP returns and the λ-derived caps that permit larger honest tickets.

Fee stance versus incumbents: one number, ρ = 2%, published, no deposit/withdrawal/turnover fees [ASSUMED]. The competitive claim is arithmetic, not adjectives: at equal flow, a toxic-flow tax removed by G2/G4 funds the price cut.

11 Implementation status and evidence

component	artifact	evidence
issuance engines	RunwayHouse (17.3kB), RunwayOptions (10.4kB) [5]	55/55 + 33/33 scenario assertions green
verifiers	MultiAttesterVerifier, MultiPriceVerifier [5]	2-of-3, floor-2, tolerance paths tested
tokenomics	EdgeDistributor, RevenueSplit, EdgeToken [5]	budget cap, mis-signed anchor, double-claim, unbond paths tested
engine	engine/ (venues, caps, pricer, quoter, attester, ledger) [6]	38/38 assertions incl. ganache interop: engine-signed quotes sell, engine attestations settle, engine merkle posts & claims on-chain
fuzzing	property fuzz, 250-op seeds 1337/42/777	251 invariant checks per seed, all green

12 Limitations (the honest page)

(i) Settlement is attester-quorum, not cryptographic trade-print proof; the L1Read verifier is the largest open item. (ii) The strike currently comes from the quoter within clamp bounds; dual-print anchored strikes (attested open as well as close) are specified but not yet shipped. (iii) κλω is a lower bound assembled from estimator assumptions; stress may invalidate the estimator faster than the breaker reacts. (iv) Equity perps trade 24/7 on one venue: off-hours is handled by cap decay plus a quoter curfew, both [ASSUMED]. (v) Regulatory: an offshore, unlicensed distribution of this product scores zero on a three-year horizon; licensing (or licensing-grade geofencing) is a launch gate, not a growth item [ASSUMED: steering position recorded by the author].

13 References and provenance ledger

[1] [CITED: Martin Galer whitepaper, docs.papertrade.xyz, fetched Oct 2026] queue-based payout bootstrap, solvency-ratio invariant, loss-emissions.

[2] [CITED: Guardian Security Review of Papertrade Protocol, final report Oct 10 2026] 42 findings; KV/H/M/L ids as used throughout this paper.

[3] [CITED: api.binance.vision snapshots fetched 2026-10-22; Bybit evaluated + dropped (geo-blocked)] spot depths/24h volumes in research/LIQUIDITY.md.

[4] [CITED: user-provided Hyperliquid screenshots, app.hyperliquid.xyz, 2026-10-20] HYPE OI/volume, market list incl. HIP-3 equity perps.

[5] [COMPUTED: contracts/ RunwayHouse.sol v3.1, RunwayOptions.sol, verifiers, EdgeDistributor.sol, RevenueSplit.sol — this repository] all on-chain behavior cited above.

[6] [COMPUTED: engine/ + test/engine.test.js — this repository] GALE engine behavior cited above.

[7+] [ASSUMED]/[RECALLED] parameters (κ, ω, ρ, tolerances, splits, curfews, market fee ranges) as marked inline.

TÓM TẮT — bản tiếng Việt

DiamondTrading là sàn turbo kiểu papertrade (nạp margin, kéo đòn bẩy đến 1000x, biết giá bust trước khi bấm, thắng thua trong 30–300 giây, thua được mint token) nhưng được dựng lại từ nền: (1) mọi khoản trả tối đa được khoá ngay khi bán — không thể vỡ nợ, không hàng đợi, không side-bucket; (2) giá settle là VWAP của lệnh khớp thật trên sàn đo được sâu nhất cho từng tài sản (Binance/Hyperliquid; HYPE đi Hyperliquid vì sâu gấp 6 lần); trần lệnh tự co giãn theo chi phí thao túng đo trực tiếp κλω; (3) mint khi thua nhưng chống farm: tính lỗ ròng theo epoch, trọng số √(lỗ × phí) × hệ số trung thành, tự hạ nhiệt khi nhà cái yếu, trần cứng 30% doanh thu thật có chữ ký; (4) phí công khai 2%. Toàn bộ đã có mã nguồn và bộ kiểm thử xanh: 55/55 + 33/33 kịch bản, 38/38 engine, fuzz 3 seed x 251 bất biến. Điểm tin cậy còn lại được khai báo rõ: bộ attester 2-trong-3 (có lối thoát hoàn tiền sau 24h), và điều kiện pháp lý để vận hành dài hạn.